

Christian Venturella

Department of Chemistry, Yale University

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Research Interests

Physics-informed machine learning for molecular and materials simulation; many-body Green’s-function methods; scalable electronic-structure theory; excited-state and response-property calculations; generative models and inverse design.

Education

Ph.D. Candidate, Theoretical Chemistry, Yale University New Haven, CT | Aug. 2022–May 2027
Advisor: Tianyu Zhu.
A.B. in Chemistry, Princeton University Princeton, NJ | Sep. 2017–May 2021

Awards and Honors

National Defense Science and Engineering Graduate (NDSEG) Fellow	2023
NSF Graduate Research Fellowship Program (GRFP) Awardee	2023
Kirkwood Fellow	2024
ACCESS Principal Investigator	2026
1.5 million CPU hours on SDSC Expanse for large-scale excited state calculations.	
Sigma Xi Research Society	2021
American Institute of Chemists Foundation Award	2021
Graduated <i>summa cum laude</i> , Princeton University	2021

Research Experience

Yale University	New Haven, CT
<i>Ph.D. Researcher, Theoretical Chemistry</i>	<i>Aug. 2022–Present</i>

- Designed physics-informed graph and graph-attention models that learn Green’s functions and self-energies from DFT orbital graphs for data-efficient ground- and excited-state predictions.
- Formulated symmetry- and polarization-aware ML representations for many-body Hamiltonians and Green’s functions, leading to publications in *Nature Computational Science* (2025) and *Journal of Chemical Theory and Computation* (2024).
- Built, curated, and validated databases of 10,000+ DFT, *GW*, and coupled-cluster calculations across molecular, nanomaterial, and solid-state systems for model development and benchmarking.
- Constructed chromophore dataset with TDDFT optical and magnetic response properties; developed excited-state descriptors to characterize charge transfer and spin-orbit effects.
- Developed low-rank Gaussian density-fitting techniques for accelerating TDDFT simulations and generative models for inverse chromophore design.
- Optimized and benchmarked low-scaling *GW* and EOM-CCSD workflows based on interacting-bath dynamical embedding theory.

Princeton University	Princeton, NJ
<i>Undergraduate Researcher, Computational Chemistry</i>	<i>Sep. 2020–Apr. 2021</i>
Senior Thesis: Modeling Nonnatural Flavoenzyme Catalysis with Molecular Dynamics and Quantum Chemical Methods	

University of Strathclyde	Glasgow, UK
<i>Visiting Undergraduate Researcher, Polymer Chemistry</i>	<i>Jun. 2019–Aug. 2019</i>

Publications

Under Review

1. C. Zhang[†], **C. Venturella**[†], E. Chen, T. Zhu. "Transferable Machine Learning of Electronic Hamiltonians with Superposition-of-Atomic-Potentials Features." *arXiv:2606.12326* (2026). Under review at *Journal of Chemical Theory and Computation*. [arXiv link](#).
2. R. Yan[†], Y. Zhu[†], Q. Gu[†], **C. Venturella**[†], H. Wang, J. Lei, T. Zhu, Y. Ye. "Anion- π Enabled Biocatalytic Asymmetric Carbanion Reactions." Under review at *Nature*.

[†] Equal contribution.

Peer-Reviewed Articles

1. J. Li, C. Hillenbrand, **C. Venturella**, E. Chen, T. Zhu. "Resolving finite-size errors in EOM-CCSD band gaps of solids with interacting-bath dynamical embedding theory." *The Journal of Chemical Physics* **165**, 094108 (2026). doi:10.1063/5.0347878.
2. **C. Venturella**, J. Li, T. Zhu. "Low-Scaling Many-Body Green's Function Calculations for Molecular Systems via Interacting-Bath Dynamical Embedding Theory." *Journal of Chemical Theory and Computation* **22**, 6599–6608 (2026). doi:10.1021/acs.jctc.6c00663.
3. **C. Venturella**, J. Li, C. Hillenbrand, X. Leyva Peralta, J. Liu, T. Zhu. "Unified Deep Learning Framework for Many-Body Quantum Chemistry via Green's Functions." *Nature Computational Science* **5**, 502–513 (2025). doi:10.1038/s43588-025-00810-z.
4. **C. Venturella**, C. Hillenbrand, J. Li, T. Zhu. "Machine Learning Many-Body Green's Functions for Molecular Excitation Spectra." *Journal of Chemical Theory and Computation* **20**, 143 (2024). doi:10.1021/acs.jctc.3c01146.

Selected Conference Presentations

ChemAI Summit, New York, NY	Summer 2026
American Chemical Society National Meeting, Atlanta, GA	Spring 2026
American Physical Society Global Physics Summit, Anaheim, CA	Spring 2025
Computational Molecular Sciences & Engineering Symposium, Yale University	Fall 2024
Yale Chemistry Symposium, Yale University	Summer 2024
American Chemical Society National Meeting, New Orleans, LA	Spring 2024

Technical Expertise

Electronic-Structure Theory: Hartree-Fock, DFT/TDDFT, MP2, coupled cluster, density fitting, response theory, Green's-function theories including *GW*.

Scientific Machine Learning: PyTorch, JAX, PyTorch Geometric, e3nn; automatic differentiation, graph neural networks, equivariant message passing.

Software and Computing: PySCF, CP2K, ORCA, Gaussian, xTB, ASE, RDKit; Python, NumPy, SciPy, pandas, scikit-learn, SQL, Matplotlib/Plotly, Linux/Bash, Git, HPC/Slurm.

Teaching and Outreach

Yale University	New Haven, CT
<i>Educational Software Developer, Demonstration Organizer</i>	<i>Apr. 19, 2026</i>
Prepared computational chemistry demonstrations for visiting middle and high school students as part of Yale Pathways Chemistry Department open house. Built educational computer game where students virtually created chromophores and simulated their colors in an OLED display.	
Yale University	New Haven, CT
<i>Teaching Fellow, Physical Chemistry (Thermodynamics)</i>	<i>September 2023–Dec. 2023</i>
Yale University	New Haven, CT
<i>Teaching Fellow, General Chemistry</i>	<i>September 2022–Jun. 2023</i>
Princeton University	Princeton, NJ
<i>Undergraduate TA, Organic Chemistry</i>	<i>Jan 2021–Jun. 2021</i>